Large-scale traffic flow reconstruction from heterogeneous mobility sensors

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Abstract

Mobile-phone data, traffic counters and GPS traces provide complementary but incomplete views of road mobility. Mobile-phone activity captures regional demand without identifying the exact roads used, traffic counters provide accurate local measurements, and GPS telemetry samples only a fraction of circulating vehicles.

We combine these heterogeneous data sources to reconstruct regional traffic flows in Emilia-Romagna, Italy. Mobile-phone origin-destination activity is transformed into trips on an OpenStreetMap road network through an agent-based simulation, producing time-dependent link-level traffic flows. Traffic counters are then used for local data assimilation of the simulated volumes by propagating residuals across nearby directed links according to simulated turning patterns. Independent Viasat GPS trajectories are held out from the reconstruction and used only to validate the simulation framework after estimating link-specific penetration rates.

The reconstructed temporal profiles are consistent with independent GPS observations on representative roads after link-specific penetration normalization. The same operational chain is integrated into a webserver for simulation management, visualization and export of reconstructed flows, and can also support regional road-safety exposure screening. This framing keeps demand reconstruction, sensor data assimilation, independent validation and downstream safety analysis as separate steps, reducing the risk of conflating heterogeneous measurements into a single observation layer.

Keywords: traffic reconstruction, mobile-phone data, traffic counters, GPS telemetry, agent-based model, data fusion

1 Introduction

Passive mobility data have changed the scale at which urban and regional transport systems can be observed. This shift is part of a broader movement toward urban computing, smart-city analytics and digital representations of urban systems [1–5]. Mobile-phone records reveal robust spatial and temporal regularities in human movement [6, 7] and have been used to estimate origin–destination (OD) matrices for transport analysis [8–10]. More generally, large geolocated data sets can support travel-demand estimation and road-use analysis when they are combined with network information and independent validation [11, 12]. These data streams do not, however, remove the central inverse problem of traffic reconstruction: demand is observed at population or areal scale, whereas congestion, emissions and infrastructure assessment depend on flows on individual road links [13–15].

A link-level traffic field is also a natural exposure denominator for road safety analysis. Crash-frequency and before–after studies have long treated traffic volume or vehicle distance travelled as a necessary reference

for interpreting accident counts, because more accidents on a road or in an hour can reflect greater exposure rather than intrinsically higher hazard [16–19]. Speed, traffic regime and road environment further modulate this relationship [20, 21]. This makes reconstructed traffic useful beyond mobility description: once validated, it can be reused to normalize aggregate accident data, detect temporal risk regimes and screen road segments whose hazard-weighted exposure is unusually high. The purpose is not to predict individual crashes, but to provide an interpretable exposure layer for regional road-safety prioritization.

Each observational layer has a different failure mode. Mobile-phone OD data provide broad coverage but do not identify the exact road sequence followed by a trip, and trip-distribution models are known to be sensitive to spatial scale and distance-decay assumptions [22–25]. Fixed traffic counters measure vehicle passages directly, yet only at instrumented locations. Floating-car or GPS telemetry is spatially richer, but it measures the vehicles present in the provider's fleet and therefore requires a road-dependent penetration normalization before comparison with total traffic [26–28]. Treating these layers as interchangeable counts can produce non-physical aggregate validations, for example by summing flows on roads that do not represent the same traffic stream. The same issue appears more generally in heterogeneous urban data, where missing values, sampling mechanisms and data governance shape what can be inferred [29–31].

This paper proposes an operational link-level reconstruction workflow that preserves the role of each data source. The contribution is not a new route-choice model, but a physically interpretable multi-layer reconstruction framework. The novelty is the integration of areal mobile-phone demand, a routable road graph, sparse fixed counters and independent GPS telemetry into a single decision-support chain while keeping their measurement meanings distinct. Mobile-phone OD activity is used to generate time-dependent demand. A directed road graph built from OpenStreetMap [32, 33] provides the routing support, using shortest-path logic derived from Dijkstra's algorithm [34] and the traffic-assignment tradition [13, 14]. Magnetic coils locally assimilate the simulated link counts where direct observations exist. Viasat GPS telemetry is kept outside the reconstruction and data-assimilation steps; it is used only as an independent validation layer for the simulation framework after estimating a link-specific penetration coefficient, so that the comparison evaluates temporal-profile consistency rather than absolute GPS fleet size. The validation links are then clustered using Ward hierarchical clustering [35, 36] to distinguish different temporal road profiles. The same reconstructed field is exposed through a dashboard/webserver so that simulation runs, data-assimilated flows, monitoring views and exports can be used by analysts without reproducing the full modelling chain in code. This operational layer is important for digital-twin and smart-city applications, where reconstructed network fields must be inspected, shared and reused for downstream questions such as traffic management and safety screening [2, 5, 37, 38].

The manuscript is organized as follows. Results and validation first present the end-to-end workflow, from the data layers to the representative validation links, then show how the reconstructed traffic field can be reused as an exposure layer for road-safety screening, and finally describe the operational webserver built around the same reconstruction chain. Methods then summarize the data, the `minimocas` traffic model, the coil-based data-assimilation step, the exposure-normalized risk analysis and the GPS validation procedure. Additional

explanations of data-layer meaning, numerical parameters, validation thresholds, road-safety descriptors and webserver architecture are reported in the Supplementary Material.

2 Results and validation

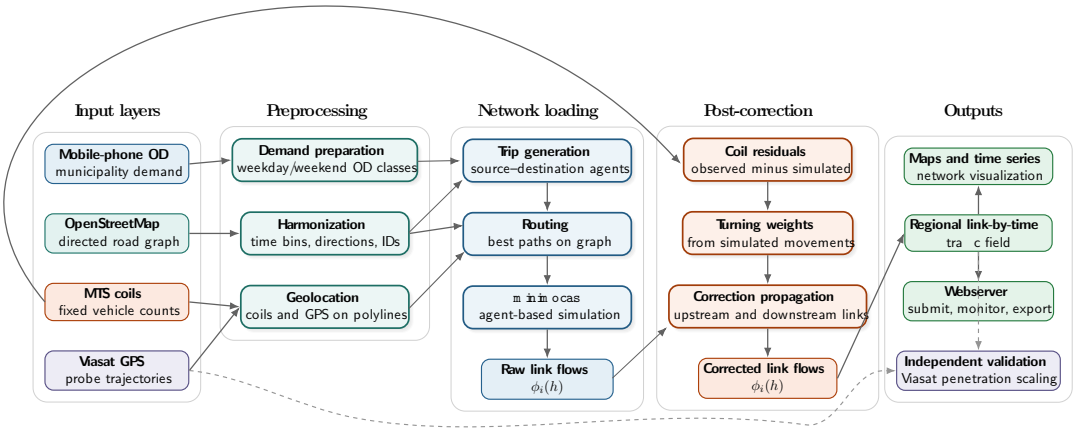

Fig. 1 End-to-end reconstruction workflow. Mobile-phone OD activity defines regional demand, OpenStreetMap provides the directed road graph, MTS coils are geolocated on road polylines and Viasat GPS traces are kept for independent validation of the simulation framework. Demand is injected into the `minimocas` agent-based simulator, producing raw link flows. Coil residuals are then used for local data assimilation of the simulated counts and propagated through adjacent directed links using simulated turning weights. The final product is a data-assimilated link-by-time traffic field that can be mapped, validated against held-out GPS traces and exposed through the webserver.

Figure 1 summarizes the logic of the reconstruction before the individual empirical results are presented. The workflow is intentionally modular: mobile-phone data define demand, the road graph defines admissible movement, coils constrain local volumes through data assimilation, and GPS traces are held out for independent validation of the simulation framework. This separation makes the data-assimilation step interpretable because the information added by fixed counters is propagated through simulated turning structure rather than through a spatial smoothing rule unrelated to traffic direction.

2.1 Observation layers and mobile-phone demand

The reconstruction is carried out on the Emilia-Romagna regional road network, whose main structure is an east–west corridor connected to dense urban areas, coastal roads and Apennine links (Fig. 2a). The mobile-phone layer is aggregated at municipality level and converted into OD activity. Weekday activity in January 2020 is strongly concentrated along the regional urban corridor, especially around Bologna and the main provincial centres (Fig. 2b). This spatial heterogeneity is the demand signal that the model must redistribute onto individual road segments, following the general idea that aggregate mobility observations must be coupled to an explicit spatial network before they can inform link-level transport questions [8, 11, 14].

The regional road traffic flow monitoring system (MTS) coils provide local vehicle counts on selected roads (Fig. 2c). Coil 282 is used as an illustrative fixed counter because it has a complete directional time series for the selected day and an unambiguous geographical record. Its 15-min profile displays the expected daily structure of a road counter: low night traffic, a morning increase and sustained afternoon activity (Fig. 2d). This combination of sparse fixed detectors and broader but sampled mobile traces is typical of traffic-state reconstruction problems [26, 28].

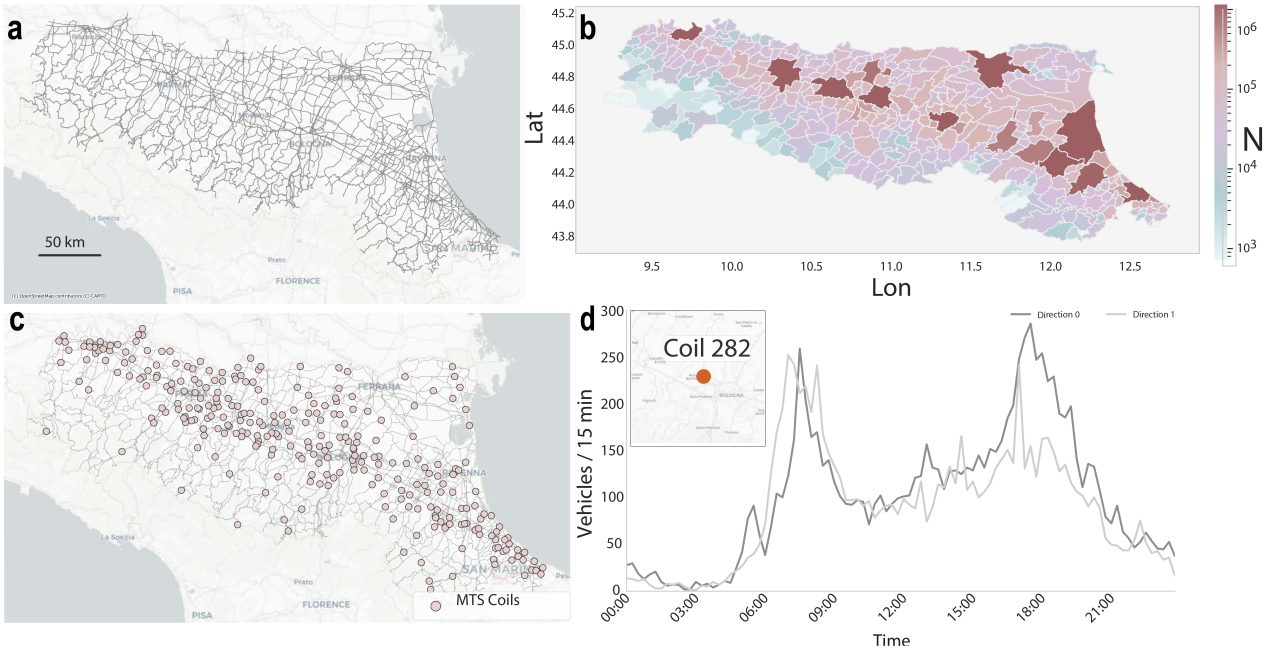

Fig. 2 Study domain and observational inputs. (a) Regional road cartography used by the model on a CartoDB Positron basemap; the scale bar corresponds to 50 km. **(b)** Weekday mobile-phone OD activity over Emilia-Romagna municipalities in January 2020; colour encodes activity N on a logarithmic scale. **(c)** Magnetic traffic-system coil locations over the regional road graph. **(d)** Detailed location of coil 282 and complete 15-min directional vehicle counts for Wednesday, 15 March 2020.

The demand layer also contains temporal and categorical information. Mean weekday activity has a sharper morning peak and a stronger late-afternoon component than weekend activity (Fig. 3a). The trip-generation classes indicate whether both endpoints are observed in the OD data or whether one or both endpoints are sampled from the modelled area (see Methods, Section 3.1). Figure 3b,c shows the mobility components used to decompose agent behaviour from the OD data. In both weekdays and weekends, the source–destination component dominates the reconstructed demand, while random-endpoint components represent residual mobility that cannot be directly linked to two observed municipalities. These classes are not arbitrary noise: they regularize weakly observed OD mass, prevent sparse low-count pairs from generating deterministic routes with false precision, preserve regional coverage and avoid an artificial collapse of weakly observed mobility onto a small number of high-count municipality pairs. Ranking municipalities by activity confirms that a small number of urban centres drive a large fraction of the demand (Fig. 3d), consistent with empirical evidence that human mobility is highly heterogeneous across places and distances [6, 22, 23].

2.2 Network reconstruction and local counter control

The simulation converts the OD demand into directional link flows. For each hour, generated agents are placed on the directed road graph, routed toward their destination and counted on every traversed polyline. The output is a link-by-time matrix $\phi_i(h)$, where i is a road polyline and h is the hour. The comparison with coil 282 shows that the simulated road associated with the counter follows the same broad daily structure as the observed series on 10 February 2020 (Fig. 4a). The map of simulated daily flows highlights the high-volume regional corridors while retaining non-zero information on the secondary network (Fig. 4b). This microscopic-to-link aggregation follows the logic of agent-based traffic simulators, where individual vehicle trajectories are used to recover network-level flow fields [39–41].

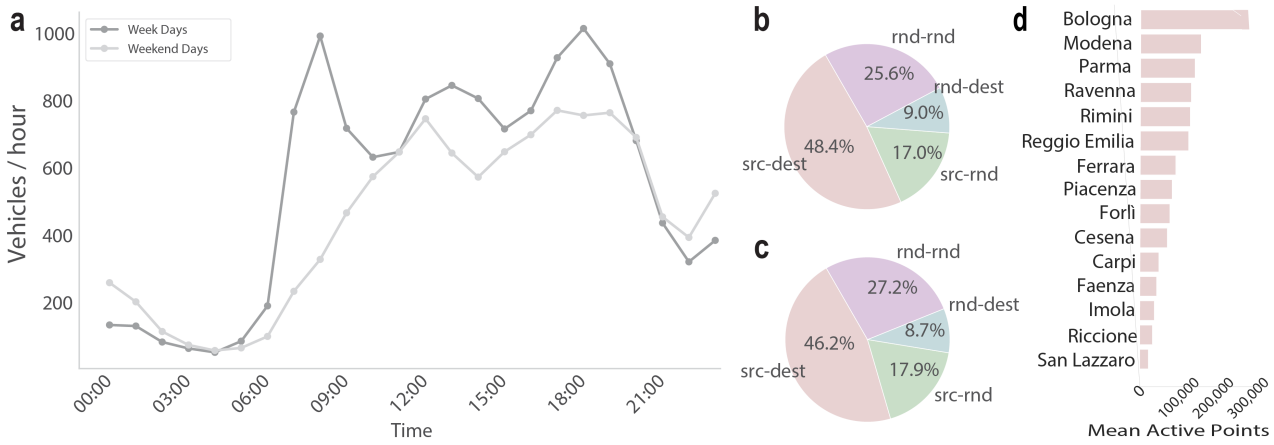

Fig. 3 Mobile-phone demand used to drive the reconstruction. (a) Mean hourly OD activity profiles for weekdays and weekend days in January 2020. **(b)** Weekday and **(c)** weekend shares of the four generation classes: source–destination (`src-dest`), source–random (`src-rnd`), random–destination (`rnd-dest`) and random–random (`rnd-rnd`). **(d)** Ranking of the most active municipalities by mean mobile activity; the plot reports the first 10 and last 10 municipalities among the top 50.

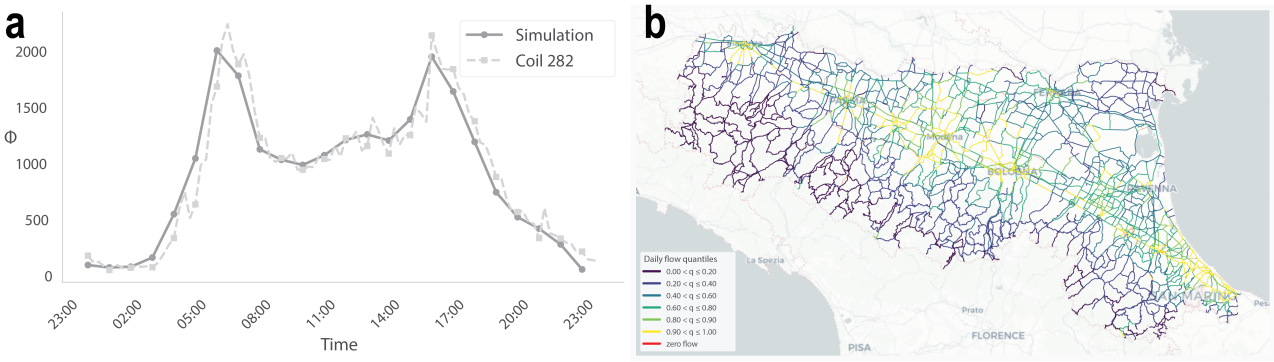

Fig. 4 Link-level reconstruction output. (a) Hourly simulated flow on the polyline associated with coil 282 compared with hourly aggregated coil observations on 15 March 2020; Φ denotes vehicle flow. **(b)** Spatial distribution of simulated daily link flows over Emilia-Romagna, expressed through flow quantile classes; yellow segments carry the highest reconstructed flows and purple segments the lowest.

2.3 Accident exposure and emergent road-risk structure

The reconstructed link-by-time traffic field is primarily a traffic product. As a downstream proof of concept, it also provides a dynamic exposure layer that can be reused for regional road-safety screening. Accident records available for Emilia-Romagna in 2022–2024 are aggregated and are not geolocated at the level of single events. For this reason, the analysis does not attempt to validate point hotspots or infer event-level causality. Instead, it asks whether the temporal and spatial structure of simulated exposure can organize aggregate accident information in a way that is useful for exposure normalization and prioritization.

We first compare the hourly accident distribution with simulated exposure profiles for weekdays and holidays (Fig. 5a). Let $A(h)$ be the share of daily accidents observed in hour h, and let $E(h)$ be the corresponding share of simulated exposure. The exposure-normalized relative risk is defined as

$$R(h) = \frac{A(h)}{E(h)}. \tag{1}$$

This normalization separates hours with many accidents because many vehicles are present from hours where the accident share is high relative to the simulated exposure. The resulting profile shows higher relative risk during night-time hours and morning transitions, while central and evening periods with more stable traffic

conditions tend to fall below the daily-average risk (Fig. 5b). Clustering the hourly normalized-risk profiles identifies distinct temporal regimes rather than a single smooth daily curve (Fig. 5c).

Across hours, accident share scales sublinearly with simulated weekday exposure, $A \sim E^{\alpha}$, with $\alpha \simeq 0.72$ and $R^2 \simeq 0.85$ (Fig. 5d). The exponent should not be read as a causal parameter. It indicates that exposure explains a large part of the temporal organization of accidents, but that accidents do not increase in direct proportion to traffic volume. This is consistent with the idea that traffic amount, traffic regime, road environment and behavioural conditions jointly shape aggregate risk.

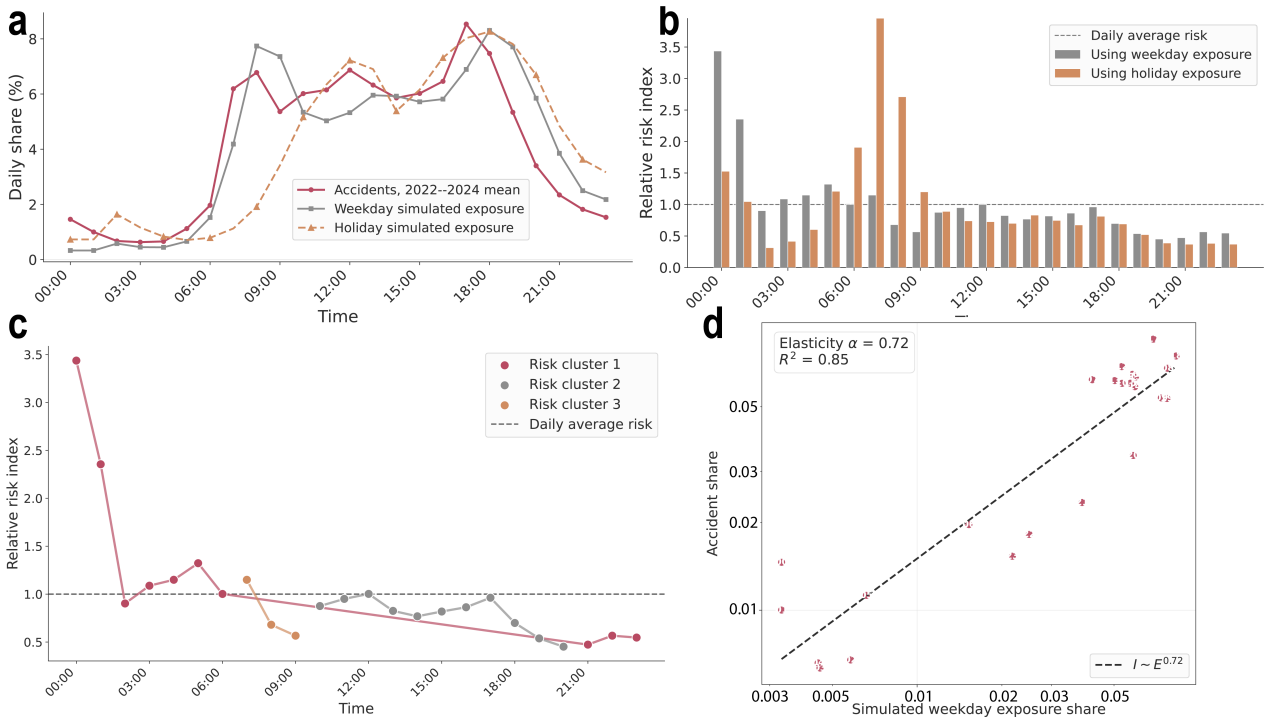

Fig. 5 Temporal risk/exposure analysis. (a) Mean hourly accident share for 2022–2024 compared with simulated weekday and holiday exposure profiles. **(b)** Exposure-normalized hourly accident intensity for weekday (grey) and holiday (orange) exposure; the dashed line marks the daily-average risk. **(c)** Clustering of hourly normalized-risk profiles into three distinct temporal risk regimes; dashed line marks the daily-average risk. **(d)** Scaling relation between simulated weekday exposure and accident share across hours (log–log axes), showing sublinear behaviour with elasticity $\alpha = 0.72$ and $R^2 = 0.85$.

The same risk profile can be projected back onto the simulated road network as hazard-weighted exposure (Fig. 6a). This produces a link-level screening layer that highlights where large simulated flows coincide with hours of elevated relative risk. The weekday–holiday comparison emphasizes how the same physical network can carry different hazard-weighted exposure patterns under different demand regimes (Fig. 6b). Ranking links by their share of total hazard-weighted exposure defines a risk centrality; the highest-ranked links are geolocated on the road map to make their spatial interpretation explicit (Fig. 6c), and their relative shares are shown as a compact bar ranking (Fig. 6d). Comparing hazard rank with exposure rank then identifies segments whose hazard is anomalously high relative to their simulated exposure (Fig. 6e). These maps are intended for stakeholder prioritization and exploratory screening, not as evidence that specific accidents occurred on the highlighted links. Additional descriptive statistics on accident totals, road environment and seasonal shares are reported in Supplementary Fig. S1.

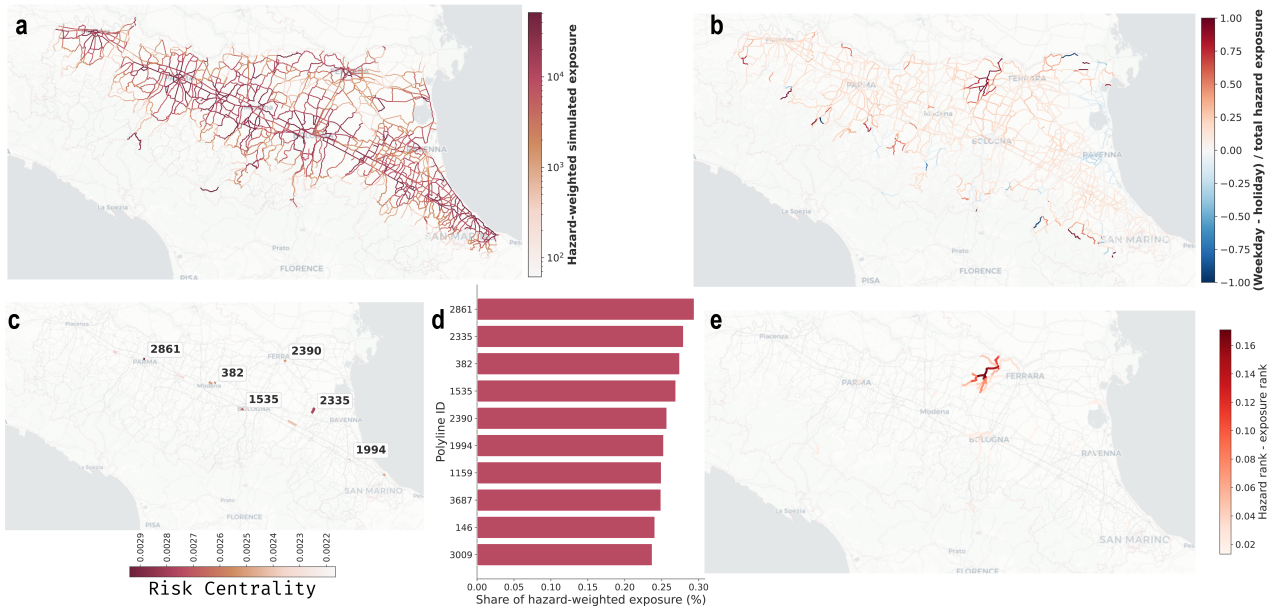

Fig. 6 Spatial organization of hazard-weighted simulated exposure. (a) Hazard-weighted exposure on the simulated regional road network, obtained by weighting link-level simulated traffic with the hourly accident-risk profile. **(b)** Relative weekday–holiday difference in hazard-weighted exposure; positive values indicate links with higher weekday hazard. **(c)** Geolocated highest-risk-centrality links; annotated polyline identifiers correspond to the top-ranked segments visible on the map. **(d)** Bar ranking of links by share of total hazard-weighted exposure; risk centrality is the fraction of regional hazard-weighted traffic carried by each link. **(e)** Road links with anomalously high hazard relative to their simulated exposure, identified by comparing hazard rank and exposure rank. Panels (a–e) represent simulated exposure weighted by aggregate temporal accident risk, not geolocated crash events.

2.4 Independent Viasat validation

Viasat GPS observations provide an independent test of the temporal profiles produced by the simulation framework. Coverage is not uniform: matched Viasat links occupy only part of the complete regional cartography, with the validation focus in the Forlì–Cesena area (Fig. 7a,d). For each matched road polyline i we estimate a statistical penetration coefficient

$$\eta_i = \frac{\sum_h \psi_i(h)}{\sum_h \phi_i(h)}, \tag{2}$$

where $\psi_i(h)$ is the raw Viasat probe count and $\phi_i(h)$ is the simulated flow. The incremented GPS profile used for validation is

$$\widetilde{\psi}_i(h) = \frac{\psi_i(h)}{\eta_i}. \tag{3}$$

This scaling aligns total volumes on each road and preserves the hourly shape of the GPS signal. Absolute GPS counts are not directly comparable with total simulated traffic because Viasat is partial-fleet telemetry and penetration varies spatially. The validation therefore evaluates whether the reconstructed daily profile is consistent with the independent probe profile after normalization, which isolates temporal consistency from fleet-size differences. It is not a validation of absolute fleet size, nor an estimate of a universal GPS market penetration. This link-specific adjustment is important when probe-vehicle coverage varies strongly across the network [26–28], and avoids treating raw GPS probe counts as complete traffic observations.

The penetration distribution is broad. In 2020, the three main penetration decades $\eta \sim 10^{-1}$, $\eta \sim 10^{-2}$ and $\eta \sim 10^{-3}$ contain 57.3%, 36.4% and 6.3% of classified roads, respectively (Fig. 7b). In 2021 the corresponding shares are 48.9%, 45.3% and 5.8% (Fig. 7e). The maps show that penetration classes are spatially interwoven,

rather than forming a single compact sampling zone (Fig. 7c,f). This motivates a validation based on individual links and profile classes rather than on a single regional aggregate.

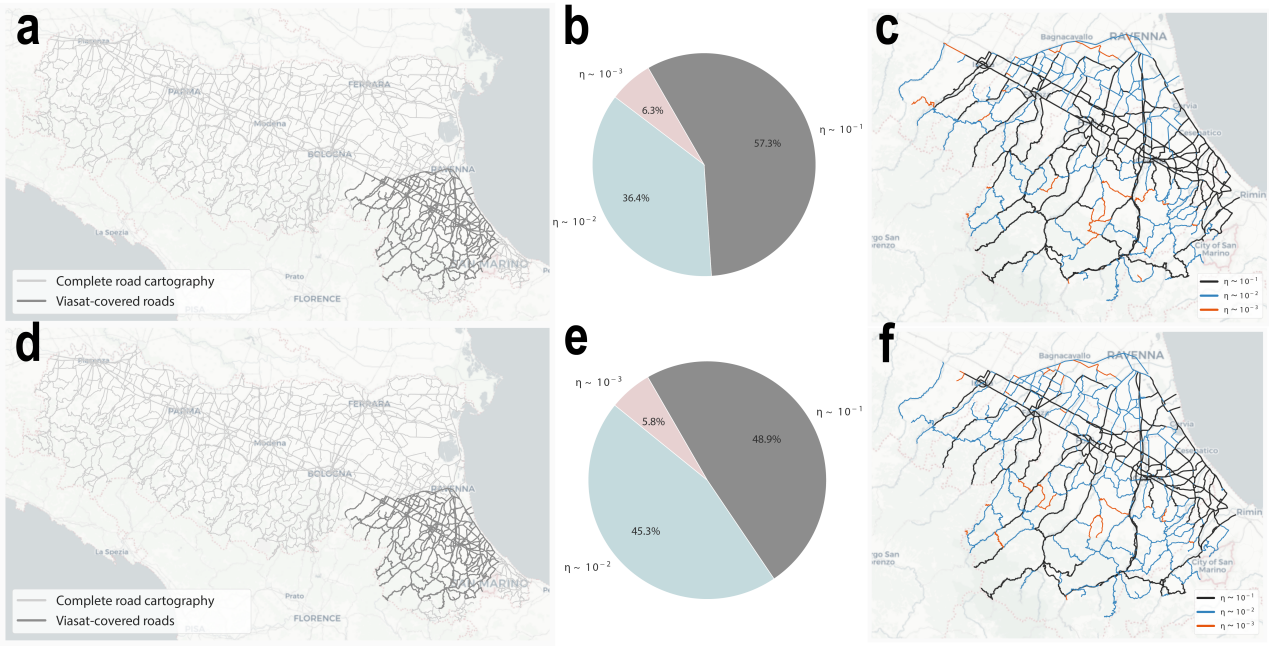

Fig. 7 Viasat coverage and statistical penetration. Top row: validation interval 10–17 February 2020. Bottom row: validation interval 28 June–4 July 2021. **(a,d)** Matched Viasat links (dark grey) over the complete regional road graph (light grey). **(b,e)** Shares of the principal link-specific penetration decades, $\eta \sim 10^{-1}$, $\eta \sim 10^{-2}$ and $\eta \sim 10^{-3}$. **(c,f)** Geographical distribution of the same penetration classes in the Forlì–Cesena validation area.

2.5 Temporal subclusters and representative roads

The most informative penetration stratum is not temporally homogeneous. In 2020, the validation archive contains 930 common simulated–Viasat links, of which 827 have usable profiles after filtering; the representative high-penetration class contains 17 candidate links (Supplementary Section S5). We therefore cluster the hourly simulated profiles of this candidate set using Ward agglomerative clustering. Cutting the dendrogram at a Ward distance of 15,000 yields three subclusters (Fig. 8a). Their spatial distribution is distinct enough to support illustrative validation examples: each cluster contributes one representative road, and the selected roads are constrained to remain geographically separated (Fig. 8b). This makes the final comparison a test of distinct temporal profiles rather than a repeated inspection of neighbouring links with nearly identical dynamics [35, 36]. The three links are not intended to be the only evidence of validation; the full matched-link statistics, candidate-filtering logic, penetration classes, mean profile discrepancies and selection outcomes are reported in Supplementary Section S5.

For each selected link, validation compares normalized hourly profiles:

$$\epsilon_i(h) = 100 \left| \frac{\phi_i(h)}{\sum_h \phi_i(h)} - \frac{\widetilde{\psi}_i(h)}{\sum_h \widetilde{\psi}_i(h)} \right|. \tag{4}$$

Polyline 104671 is kept as the charcoal representative because it is close to the urbanized Forlimpopoli area and therefore has a clear geographical interpretation. The remaining two representatives are selected algorithmically from the other clusters by minimizing the mean normalized-profile discrepancy subject to a maximum mean

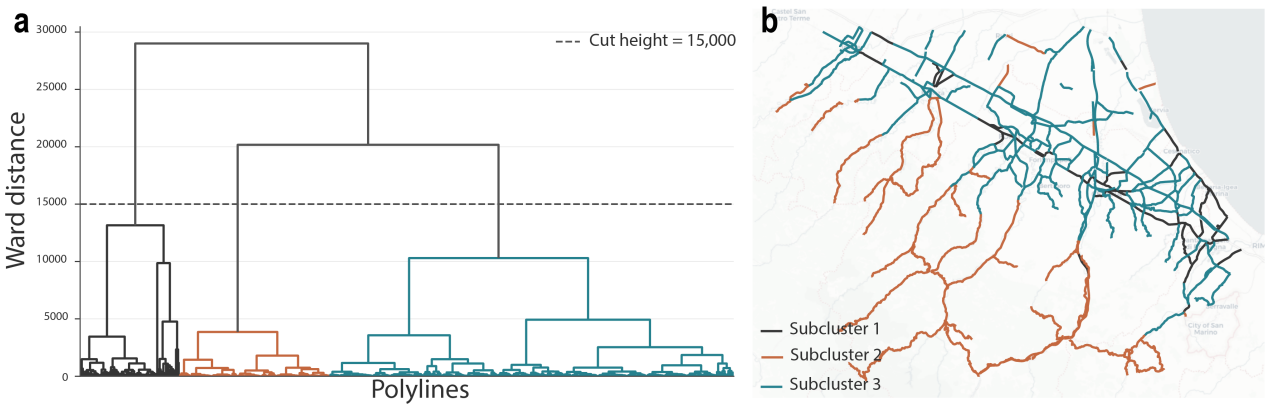

Fig. 8 Subclustering of the representative GPS-penetration stratum in 2020. (a) Ward dendrogram of hourly link-flow profiles; the dashed line marks the adopted cut at 15,000. **(b)** Spatial location of the three resulting subclusters, using the same colours as the dendrogram.

error of 0.80% and a minimum pairwise distance of 40 km. These thresholds are reporting constraints rather than fitted parameters or optimization targets: the error limit keeps the examples within the sub-percent validation regime, whereas the distance limit prevents selecting three visually redundant neighbouring roads. The resulting links are 187204 and 132443. Their mean discrepancies are 0.53%, 0.76% and 0.61%, respectively, and the minimum pairwise separation is 41.9 km (Fig. 9). The visual agreement of the daily profiles illustrates the reconstructed temporal shape on roads with different cluster membership and different positions in the validation area; broader validation distributions are reported in Supplementary Section S5.

2.6 Operational webserver

The reconstruction workflow was also implemented as a webserver so that the validated reconstruction chain can be deployed as an operational decision-support service rather than only as an offline analysis script. The dashboard is therefore an interface to the same modelling, data-assimilation and export logic described above, not a separate software contribution. It separates the browser-facing interface from the simulation backend. Through the interface, an authenticated user chooses a simulation interval and submits a run request. The backend generates the corresponding model configuration, launches the `minimocas` simulation asynchronously, monitors progress and applies the coil-based data-assimilation step before returning the reconstructed link-flow table. This design makes the assumptions, inputs and outputs reproducible while allowing stakeholders to interact with reconstructed traffic fields through maps and tables. It follows the smart-city idea that mobility models become more useful when they are exposed as repeatable analytical services, while still requiring transparent assumptions and validation [2–4, 31].

This implementation makes three practical outputs available to non-developer users. First, completed simulations are stored with metadata such as user, start time, runtime, completion and status, so previous analyses can be reopened. Second, data-assimilated flows can be exported as a table indexed by road polyline and time. Third, the road network can be visualized interactively: single simulation details combine time-series plots with a map of selected links, while aggregate flux maps colour all polylines by reconstructed flow quantiles. The same dashboard includes monitoring views for the MTS coil data, including station maps, time series,

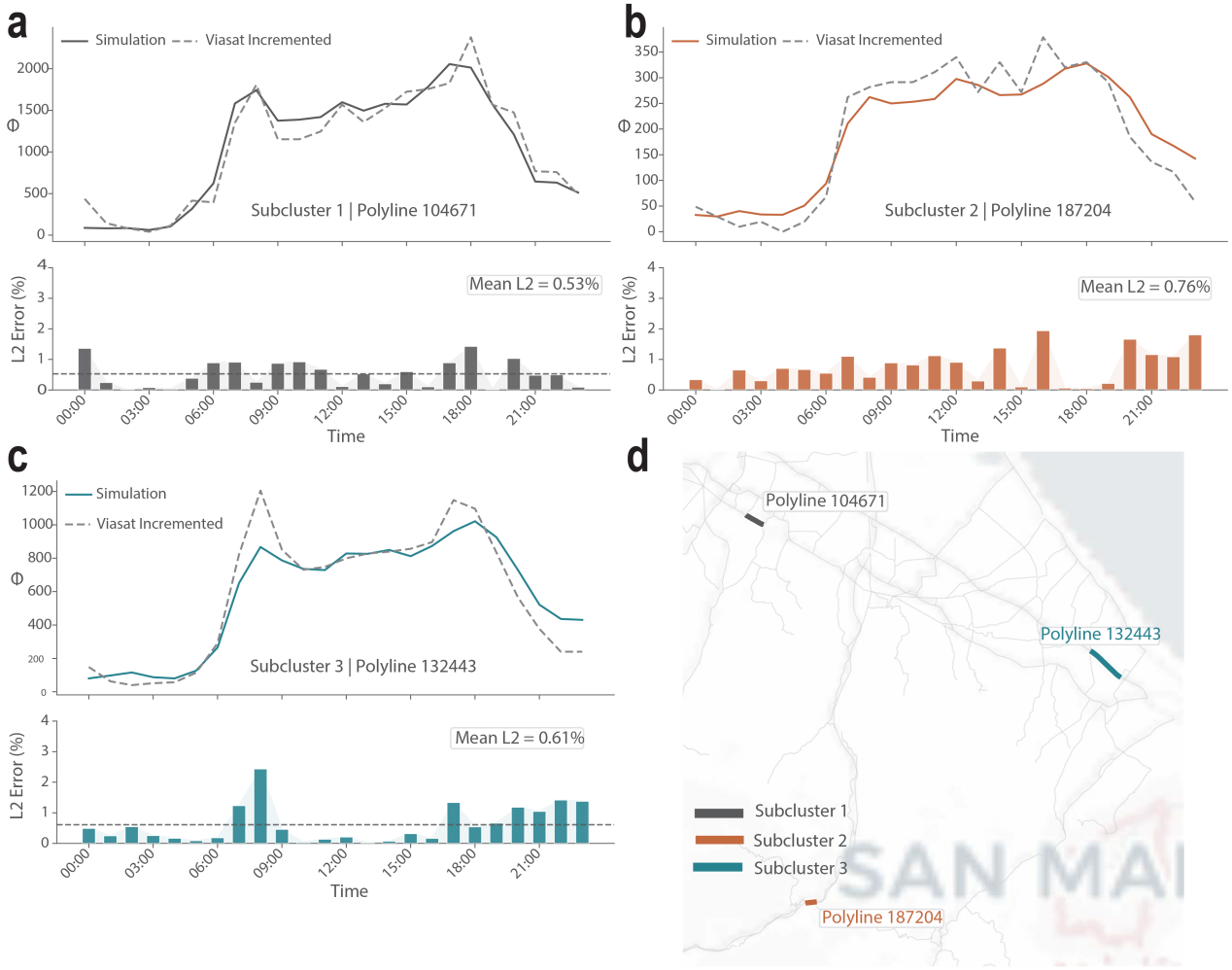

Fig. 9 Validation on spatially distributed representative polylines, 10–17 February 2020. (a–c) Mean simulated hourly flow and incremented Viasat profile, with normalized-profile discrepancy below. **(a)** Polyline 104671, mean error 0.53%. **(b)** Polyline 187204, mean error 0.76%. **(c)** Polyline 132443, mean error 0.61%. **(d)** CartoDB Positron map of the three selected polylines, coloured as in Fig. 8.

rankings, clusters and anomaly diagnostics. Supplementary Section S7 describes the webserver components, user workflow and API logic in tabular form.

3 Methods

3.1 Data sources and preprocessing

The road support is a routable regional graph derived from OpenStreetMap cartography. Road polylines are represented by local and cartographic identifiers, front and tail nodes, length and direction metadata. A preprocessed cartography file defines the node–polyline adjacency used both by the simulator and by the data-assimilation algorithm. The preprocessing follows standard graph-based principles for converting detailed cartography into a transport network that is simple enough to simulate while preserving the topological relations relevant to routing [32, 33, 42–46].

Mobile-phone mobility is aggregated to municipality-level OD activity. The January 2020 records are separated into weekdays and weekend days and into four generation classes: observed source–observed destination (`src-dest`), observed source–random destination (`src-rnd`), random source–observed destination (`rnd-dest`) and random source–random destination (`rnd-rnd`). These classes determine how origins and destinations are

sampled when demand is injected into the simulation. The construction keeps the empirical OD component explicit and uses distance-aware random endpoints only for residual demand that cannot be assigned to two observed municipalities. The random-endpoint classes therefore act as a controlled residual mobility mechanism: they regularize sparse OD mass, preserve coverage of the regional graph and avoid forcing weakly observed flows into deterministic routes with spurious precision. This reflects the sensitivity of trip-distribution models to spatial support and missing observations [23, 24, 29, 30, 47].

MTS coil data are matched to road polylines through barrier metadata. A direction-pivot table maps registry directions to cartographic directions (`IN`/`OUT`). When direct coil observations are missing for the simulation interval, the data-assimilation code fills them using seasonal weekday averages, combining the most recent year and previous years with inverse standard-deviation weights. Viasat GPS data are independently map-matched to road polylines and are used only to validate the simulation framework; they are not used to assimilate or tune simulated flows. The meaning, scale and role of each data layer are summarized in Supplementary Section S1.

3.2 The `minimocas` traffic model

`minimocas` is an agent-based traffic simulator in which each vehicle is represented by an agent called `"pawn"`. A pawn has a current polyline, a position along the polyline, a speed, a destination and a status. Simulation time advances with time step Δt; in the Emilia-Romagna configuration the code supports sub-hourly evolution and hourly camera dumps. During each time step, active pawns move along their current polyline. If they reach a node, the next polyline is chosen from the best-path matrix for the current destination, with an optional probability of missing the best path. The best-path matrix is computed on the weighted graph with a Dijkstra-style engine and respects no-turn constraints. The objective is not behavioural route-choice realism or dynamic congestion assignment. Shortest-path routing, the absence of adaptive rerouting and the simplified congestion rule are deliberate choices that favour stable reconstruction of aggregate directional flow fields from uncertain demand. The model is deliberately simpler than full microscopic platforms such as MATSim and SUMO [39, 40], but it shares their agent-based representation of trips and network loading.

Sources inject new pawns according to time-varying creation rates. The daily creation-rate vector is indexed by simulation time, allowing the mobile-phone hourly demand profile to modulate the number of generated trips. Depending on the generation class, the model can use observed origins and destinations, draw one missing endpoint conditionally, or draw both endpoints from the regional node set. The selected OD pair is converted into an initial directed polyline and a destination node; the pawn then follows the network until the trip is completed or killed by model rules.

The simulator records two forms of traffic output. Barrier counters count crossings at coil-like virtual barriers. Unique polyline output records, for each dump interval and each direction, the number of distinct pawn identifiers present on a polyline. This second output is the link-level quantity used in the reconstructed flow maps and in the subsequent data-assimilation step.

The model also updates link velocities from simulated density. For each polyline i, density is the number of pawns on the link divided by its length. The code imposes a critical-density rule: above the critical density the

link speed is set to a critical speed; below it, the nominal polyline speed is linearly reduced with density. The link flow used internally is the product of density and updated speed. This places the implementation within the broader family of traffic-flow and car-following models that link microscopic motion to macroscopic density, speed and flow [41, 48–55]. Supplementary Section S2 provides the run inventory and source-class counts, while Supplementary Section S3 describes the model-state variables.

3.3 Coil-based data assimilation

The data-assimilation procedure implemented in the framework adjusts the simulated unique-link matrix using MTS coil residuals. Let $u_i(h)$ denote the simulated unique count on the polyline associated with coil i, $b_i(h)$ the corresponding synthetic barrier count, and $c_i(h)$ the observed coil count after direction mapping and temporal harmonization. The code first estimates the relation between barrier and unique counts,

$$r_i(h) = \frac{b_i(h)}{\max(u_i(h), 1)}, \tag{5}$$

and computes a relative residual in barrier space,

$$\Delta_i(h) = \frac{c_i(h) r_i(h) - b_i(h)}{\max(b_i(h), 1)}. \tag{6}$$

This residual is converted into an additive assimilation increment on the corresponding directed road count. Negative assimilated counts are clipped to zero in the final output. The procedure is a local data-assimilation step: fixed counters constrain the simulated field where observations are available, while the simulation supplies the spatial structure needed to propagate information to nearby unobserved links [26, 28]. It should not be interpreted as a full dynamic traffic-assignment calibration of the entire network.

The assimilation increment is not applied only at the coil polyline. It is propagated recursively through adjacent directed links using turning weights inferred from the simulated unique counts. For an incoming directed link (n_1, n_2) and an outgoing neighbour (n_2, n), the forward weight is the share of simulated flow continuing toward n among all admissible outgoing links. If the outgoing simulated counts are missing, the weight is replaced by a uniform split across admissible alternatives. A symmetric backward recursion distributes the same local assimilation increment upstream. The recursion stops when the absolute propagated increment falls below one vehicle, when depth exceeds five links, when another coil polyline is reached, or when a no-turn or patched-closed polyline would be crossed. The data-assimilated table is written in the same directional unique-polyline format as the simulator output. The propagation depth, one-vehicle stopping threshold and fallback uniform turning weights are operational parameters used to keep the data assimilation local and stable when simulated turning counts are sparse. Poorly instrumented areas therefore remain more uncertain than roads close to reliable counters. Supplementary Section S4 reports the data-assimilation rules, including the rationale for the one-vehicle stopping threshold and five-link recursion limit, and the coil-282 summary used for the local comparison.

3.4 Accident exposure and hazard-weighted traffic

The road-safety analysis uses regional accident records for 2022–2024 as aggregated counts. The records are suitable for describing temporal distributions and broad road-environment composition, but they are not used as geolocated single-event points. Consequently, the method does not geolocate individual crashes, predict individual accidents or calibrate observed crash hotspots. It combines aggregate accident timing with the simulated traffic field to construct exposure-normalized and hazard-weighted screening indicators.

Let $\phi_i(h)$ be the simulated flow on link i during hour h. Hourly exposure is obtained by summing simulated traffic over the road network,

$$E(h) = \frac{\sum_i \phi_i(h)}{\sum_{h'} \sum_i \phi_i(h')}, \tag{7}$$

so that $E(h)$ is a daily share. Accident counts are normalized in the same way,

$$A(h) = \frac{a(h)}{\sum_{h'} a(h')}, \tag{8}$$

where $a(h)$ is the number of accidents in hour h. The exposure-normalized hourly risk is then Eq. (1). Separate weekday and holiday exposure profiles can be used in the denominator when the goal is to compare temporal regimes.

For spatial screening, the hourly risk profile is projected onto each link through hazard-weighted exposure,

$$H_i = \sum_h \phi_i(h) R(h). \tag{9}$$

This quantity is high when a link carries large simulated traffic during hours with elevated aggregate relative risk. The weekday–holiday contrast is computed as

$$D_i = \frac{H_i^{\text{weekday}} - H_i^{\text{holiday}}}{H_i^{\text{weekday}} + H_i^{\text{holiday}}}, \tag{10}$$

with the denominator evaluated only where the total hazard-weighted exposure is positive. Link risk centrality is

$$C_i = \frac{H_i}{\sum_j H_j}, \tag{11}$$

and measures the share of total hazard-weighted exposure carried by link i. Finally, anomalous hazard relative to exposure is screened by comparing ranks,

$$S_i = \text{rank}(H_i) - \text{rank}(E_i), \tag{12}$$

where $E_i = \sum_h \phi_i(h)$ is total simulated link exposure and ranks are oriented so that larger values correspond to higher hazard or exposure. Large positive deviations indicate links whose hazard-weighted exposure rank is

high compared with their baseline exposure rank. These indicators support screening and prioritization, but they are not interpreted as proof of causal risk or as geolocation of individual accidents.

3.5 GPS validation and clustering

Validation uses Viasat only after matching GPS roads with simulated polylines. The GPS data are not used to generate demand, assimilate coil residuals, tune the simulation or select model parameters. They provide an external check of the temporal profiles produced by the reconstruction framework. For every matched road, the link-specific penetration coefficient in Eq. (2) rescales the GPS series so that total volume on that road is aligned before profile comparison. Because this scaling uses each road separately, it tests the temporal profile on that physical link rather than a region-wide sum or a universal market-penetration factor. The normalized-profile discrepancy in Eq. (4) is used as the hourly validation diagnostic because absolute probe counts are not total traffic counts: Viasat observes a partial fleet, and its penetration varies across links. Normalization isolates whether the simulated and observed profiles rise, peak and decline at comparable hours, which is the validation target for the reconstructed temporal field. This is a conservative use of probe data: the GPS traces validate shape and timing after penetration adjustment, but they are not treated as complete traffic counts and are not merged back into the calibration or data-assimilation stages [12, 27, 28].

To choose interpretable examples for the paper, links in the representative penetration class are clustered by their simulated hourly profiles using Ward linkage and Euclidean distance. The dendrogram is cut into three subclusters. Polyline 104671 is fixed as the charcoal representative because of its urban location near Forlimpopoli. Representatives from the other two subclusters are selected by minimizing mean normalized-profile error while enforcing a maximum mean error of 0.80% and a minimum inter-link separation of 40 km. This selection avoids visually redundant neighbouring examples and keeps one validation road for each temporal subcluster. The examples are illustrative representatives of temporal and geographical regimes, not the sole validation evidence. Extended validation tables, threshold rationale, candidate-link counts and representative-link metrics are reported in Supplementary Section S5.

4 Discussion

The main methodological contribution is the construction of an operational reconstruction chain in which heterogeneous mobility sensors keep explicit semantic roles. Mobile-phone records define the regional demand signal; the road graph and simulator perform the network assignment; coils provide local data assimilation where direct counts exist; and Viasat is reserved for independent temporal-profile validation of the simulation framework. Each observational layer preserves its original measurement meaning, and the workflow explicitly separates demand generation, routing, data assimilation and validation. This separation is important because each layer has a different sampling mechanism and a different spatial support. It also follows a central lesson of urban analytics: large mobility data sets are powerful when their observation process is kept visible, not when they are treated as complete observations [1, 3, 4, 31].

The validation results show that the simulation framework can produce reconstructed daily profiles consistent with held-out, incremented GPS profiles on illustrative roads from three temporal subclusters. At the same time, the penetration maps and supplementary validation tables make clear that validation strength is not uniform over the entire cartography. The workflow is therefore best interpreted as a transparent data-fusion and validation pipeline, not as proof that every low-penetration road is reconstructed with the same accuracy. This validation focuses on temporal-profile consistency after penetration normalization, rather than on absolute fleet-size estimation. The latter would require additional assumptions and data sources, such as fleet composition, vehicle-kilometres travelled, fuel sales or regional population. The current strategy is more conservative: it tests whether the temporal dynamics produced by the reconstruction framework are consistent with an independent probe-vehicle sample after link-specific normalization, but it does not estimate a universal GPS market share or claim absolute calibration of all links.

The practical value of the workflow is that it produces a road-level traffic representation from data sources that are normally used separately. This output can be used to identify corridors with high reconstructed flows, compare weekday and weekend demand scenarios, inspect the temporal profile of specific roads, support local traffic-management decisions and provide an exposure layer for downstream analyses such as emissions, noise, road-safety screening or infrastructure stress. Because the reconstructed quantity is indexed by road polyline and time, it can also be aggregated after the fact to municipalities, provinces, corridors or user-defined areas without re-running the model. This is the type of intermediate mobility layer needed by data-driven planning tools, accessibility studies and urban digital-twin applications [2, 5, 37, 38, 56].

The accident analysis illustrates a downstream use of the reconstructed field rather than a second calibration target. The available 2022–2024 accident data are aggregate observations, so the paper does not geolocate individual crashes, interpret the maps as observed crash hotspots or claim to predict individual accidents. Instead, simulated traffic is used as a dynamic exposure denominator. Normalizing accidents by exposure reveals temporal risk regimes that are not visible from accident counts alone: night-time and transition hours can have elevated relative risk even when their absolute counts are moderate, whereas high-volume central hours can have lower relative risk because exposure is also high.

The observed scaling $A \sim E^{0.72}$ reinforces the same interpretation. Exposure accounts for a large fraction of the temporal accident structure, but the sublinear exponent indicates that accident share does not rise proportionally with simulated traffic share. Road environment, speed regime, visibility, behavioural conditions and reporting processes may all contribute to the remaining variation. The spatial hazard-weighted maps should therefore be read as screening and prioritization tools: they identify links whose simulated traffic occurs during more hazardous hours, and links whose hazard rank is high relative to their exposure rank, but they do not establish causal mechanisms or event-level accident locations.

The webserver implementation is important because it operationalizes the validated reconstruction workflow without changing its scientific assumptions. It turns the offline chain into a repeatable service: a user can choose a time window, launch the simulation, monitor progress, download data-assimilated flows and visualize the result on the road graph. This makes the workflow suitable for reproducible scenario analysis, operational

reporting and communication with stakeholders who need interpretable traffic fields as maps and tables rather than source code. The dashboard also keeps the MTS monitoring layer close to the simulation layer, allowing counter data quality, anomalies and data-assimilation behaviour to be inspected in the same environment. The detailed service architecture is summarized in Supplementary Section S7.

The main limitation of the present implementation is the seasonal support of the OD matrix. The demand layer used here is based on winter mobility observations, and therefore represents a specific seasonal configuration of regional mobility. Access to multiple OD matrices, ideally derived from different data sources and covering different seasons, would allow the same framework to reproduce seasonal changes in mobility demand more explicitly and to test how robust the reconstructed link-flow field is under different demand regimes.

A second limitation is methodological and operational. The simulator is intentionally simplified to keep regional runs compatible with webserver deployment and interactive stakeholder use. It does not implement a full dynamic congestion mechanism or adaptive traffic assignment; instead, it uses shortest-path network loading together with a free-flow/threshold velocity representation. This choice is coherent with the main target of the study, which is the reconstruction of aggregate directional flows on a large extra-urban road network rather than the detailed modelling of urban congestion waves. Within this setting, the coil-based data-assimilation step is the mechanism that anchors simulated flows to local observations while preserving feasible runtimes for repeated operational analyses.

Future work should therefore focus on extending the demand basis and testing the model under richer seasonal and multi-source OD scenarios. On the operational side, the webserver could be extended with explicit scenario controls, seasonal demand selection, user-defined regions of interest and automatic reporting of the OD and sensor layers used in each run.

5 Conclusions

This work presents a regional-scale workflow for reconstructing road traffic flows by combining mobile-phone activity, fixed traffic counters, OpenStreetMap cartography and GPS telemetry. The contribution is not a new routing algorithm, but an interpretable multi-layer reconstruction framework in which demand generation, network loading, local data assimilation and independent validation remain explicitly separated. The proposed chain integrates heterogeneous mobility observations into a coherent time-dependent traffic field at the road-link scale over a large regional network.

Held-out Viasat telemetry validates the simulation framework by showing strong consistency between reconstructed and GPS-derived hourly profiles on representative roads from different temporal mobility regimes after link-specific penetration normalization. This should be interpreted as a validation of reconstructed temporal dynamics rather than as a claim of complete absolute calibration across the entire network. The results nevertheless show that large-scale mobility patterns can be reconstructed with sufficient temporal and spatial coherence to support downstream operational analyses.

A central result of the study is that the reconstructed traffic field can also be reused as a dynamic exposure layer for regional road-safety assessment. By combining simulated road exposure with aggregate accident

statistics, the framework makes it possible to distinguish highly trafficked corridors from road segments exhibiting disproportionately high hazard relative to their estimated exposure. The analysis reveals marked temporal risk heterogeneity across the day and highlights the existence of road segments whose accident intensity cannot be explained solely by traffic volume. This transforms the reconstructed mobility field into a practical screening tool for identifying potentially critical infrastructures and prioritizing monitoring or intervention strategies.

Beyond scientific analysis, the workflow was designed with stakeholder-oriented applications in mind. The dashboard implementation supports simulation management, monitoring, visualization and export of reconstructed flows, enabling regional analysts to inspect mobility scenarios without directly interacting with the modelling pipeline. Potential applications include congestion screening, infrastructure stress assessment, environmental-impact analysis, exposure-based safety prioritization and digital-twin support for regional mobility management.

Future developments should incorporate uncertainty quantification, systematic sensitivity analyses for the data-assimilation procedure and broader validation campaigns across additional territories and temporal conditions. Further extensions could integrate weather conditions, road characteristics and near-real-time mobility observations to support predictive hazard assessment and adaptive traffic-management strategies.

Supplementary Material

Supplementary Material provides tabular documentation of the data layers, simulation runs, `minimocas` state variables, data-assimilation rules, validation metrics, accident descriptors and webserver architecture referenced throughout the main text. Supplementary Fig. S1 reports additional descriptive statistics on accident totals, road environment and seasonal shares, and Supplementary Section S8 summarizes reproducibility notes for the reported counts, thresholds and operational workflow.

Availability of data and materials

Processed figure inputs and the code required to regenerate the analyses are available in the project repository. Public traffic-counter data from the Emilia-Romagna Region are available through the MTS online mobility platform at https://mobilita.regione.emilia-romagna.it/servizi/flussi-online. Road-accident statistics are publicly available from ISTAT at https://www.istat.it/tag/incidenti-stradali/. Mobile-phone and Viasat GPS datasets used in this study are proprietary and cannot be publicly distributed.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

FB conceived part of the original study design, developed the reconstruction workflow, performed the analyses and prepared the first manuscript draft. CM and AF contributed to the development of the modelling framework. AB contributed to the study design, interpretation of the modelling results and manuscript revision. All authors read and approved the final manuscript.

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